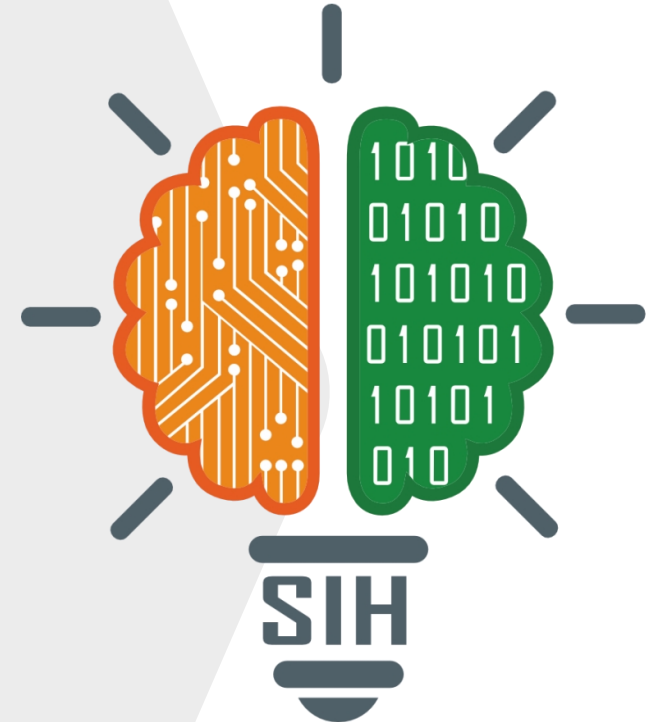


## TITLE PAGE

- **Problem Statement ID – 8-L**
- **Problem Statement Title-Smart Waste Management and citizen reporting platform**
- **Theme- Digital Solutions for Smart cities**
- **PS Category- Smart Governance**
- **Team ID- 114\_Code\_Craft**
- **Team Name (Registered on portal)- Code Craft**



# SWAACHX — FROM REPORT TO SMART ACTION

## ❖ Proposed Solution

### REPORT

- Citizens report waste issues using location, photo, voice or text.

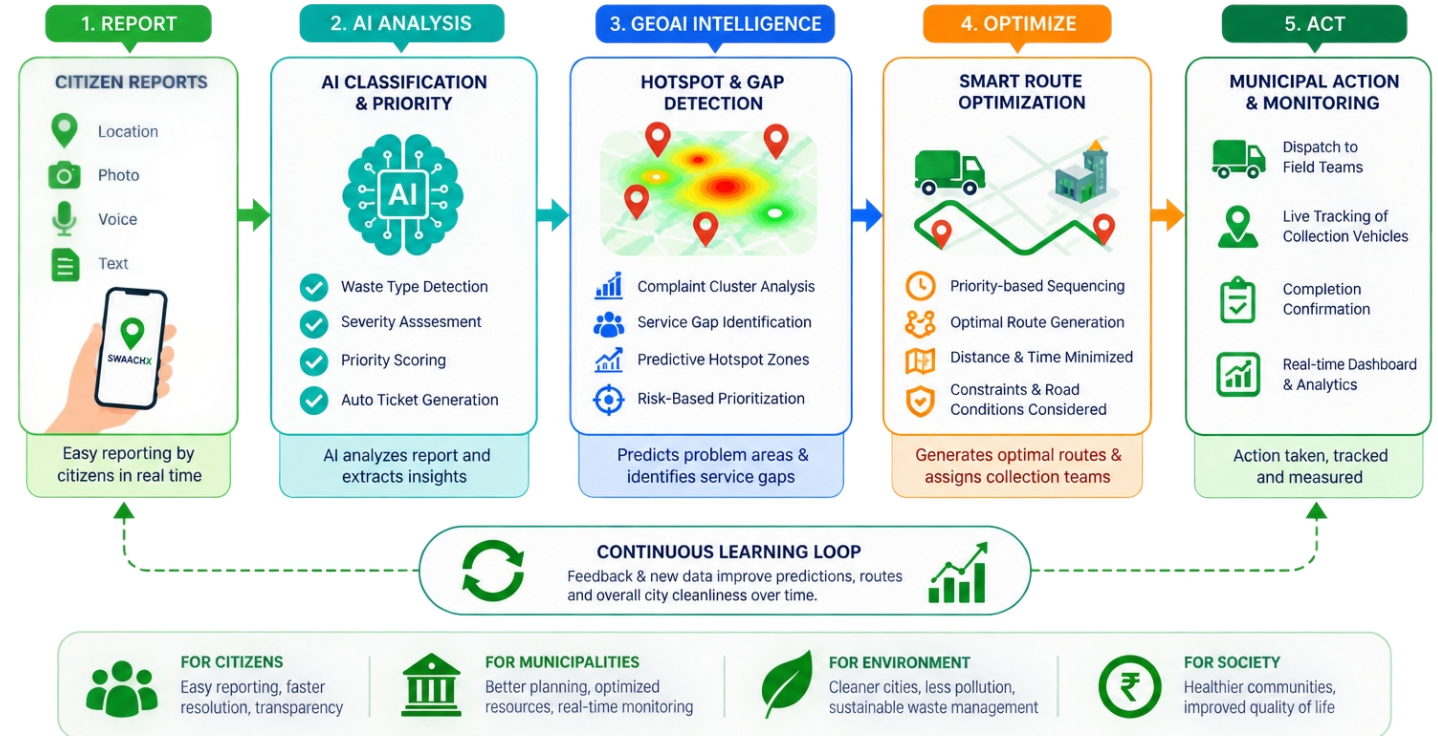
### PREDICT

- AI +GeoAI identify waste hotspots and service gaps

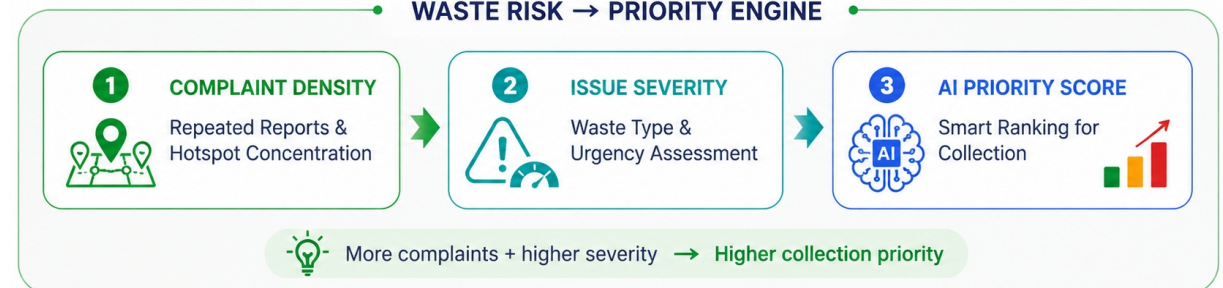
### OPTIMIZE

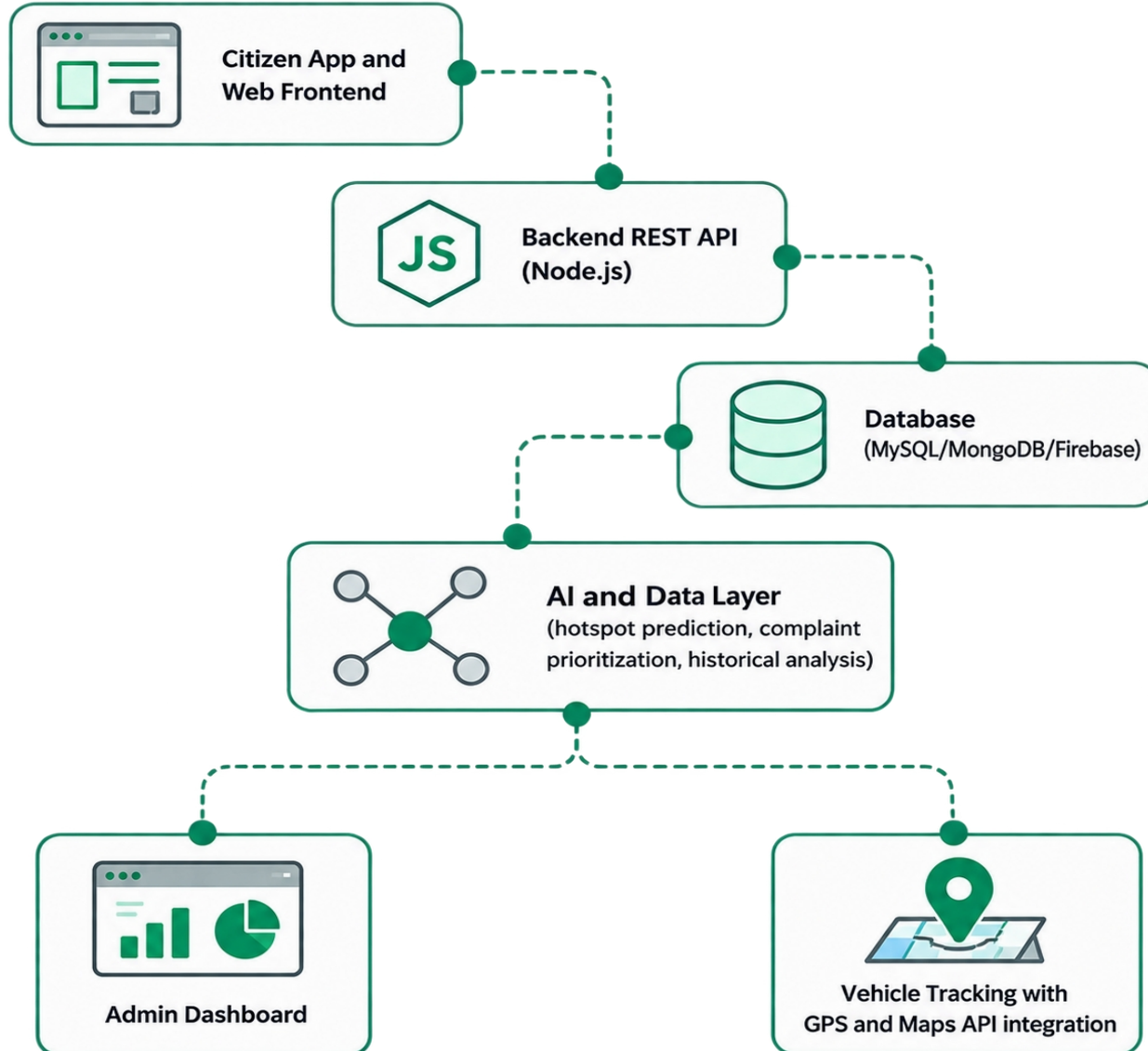
- Smart routing enables priority-based and efficient collection.

#### FROM CITIZEN REPORT TO SMART ACTION



#### WASTE RISK → PRIORITY ENGINE





## ❖ Prototype

- Citizen waste reporting
- Status tracking
- Collecting vehicle tracking
- Admin interface

## ❖ Tech Stack

- Frontend: React+vite
- Backend: Node.js /Rest API
- Database: MySQL/MongoDB
- Maps: Google maps

## ❖ Planned Integrations

- GPS / vehicle tracking
- Municipal Complaint Systems
- Map APIs
- IoT sensors

## ❖ Feasibility Analysis

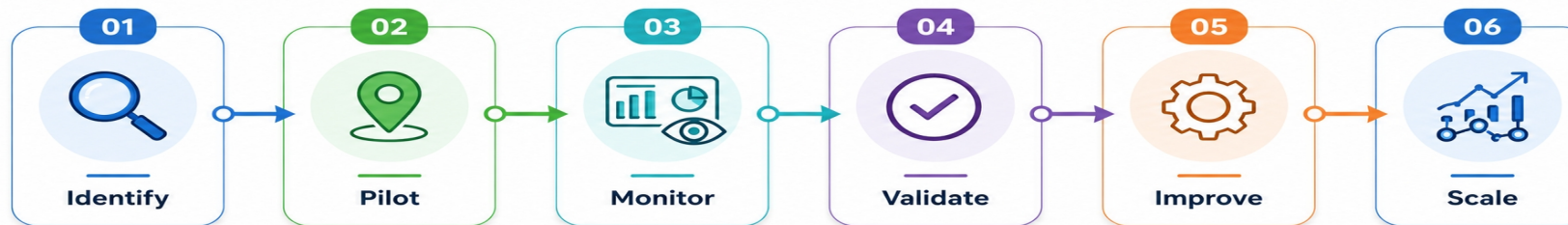
- Uses proven technologies like GPS, maps, cloud and AI/ML.
- Cost-effective with open-source tools and existing infrastructure.
- Scalable from selected wards to city-wide implementation.

## ❖ Potential Challenges & Risks

- Inaccurate or duplicate reports may affect data quality.
- Limited historical data can reduce prediction accuracy.
- Network, GPS and adoption issues may affect reliability.

## ❖ Strategies to Overcome

- Verify reports using AI-based and rule-based validation.
- Improve predictions continuously as more data is collected.
- Provide simple interfaces and offline support for accessibility.



## ❖ Potential Impact on Target Audience & Benefits

### Citizens

- Easy Reporting — Report waste issues quickly.
- Transparency — Track complaints and resolution status.
- Participation — Encourages citizens to contribute to cleaner communities.

### Municipal Authorities

- Faster Response — Identify and prioritize waste hotspots.
- Higher Efficiency — Optimize collection routes and resources.
- Better Decisions — Use real-time data for smarter planning.

### Community & Environment

- Cleaner Surroundings — Reduce unmanaged waste accumulation.
- Lower Resource Use — Minimize unnecessary collection trips.
- Sustainable Cities — Enable smarter and scalable waste management.



- Key sources and technical documentation used in developing SWACCHX

## Domain and Government Research

01 — **Swachh Bharat Mission** – Urban  
Urban sanitation & waste-management  
framework

 [sbmurban.org](https://sbmurban.org)

02 — **Ministry of Housing & Urban Affairs**  
Urban development & sanitation resources

 [mohua.gov.in](https://mohua.gov.in)

03 — **Central Pollution Control Board**  
Solid waste-management & environmental  
resources

 [cpcb.nic.in](https://cpcb.nic.in)

## Technical References

04 — **Google Maps Platform**

Maps, geolocation & routing documentation

 [developers.google.com/maps](https://developers.google.com/maps)

05 — **React Documentation**

Frontend architecture & development

 [react.dev](https://react.dev)

06 — **OpenStreetMap**

Open geographic & mapping data

 [openstreetmap.org](https://openstreetmap.org)